

Bus@Belimo

Bus Basics B1



Modbus RTU



Agenda Tuesday

Bus@Belimo Bus Basics B1



Analog ⇨ Digital

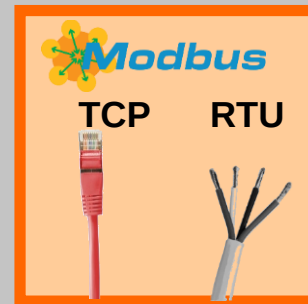
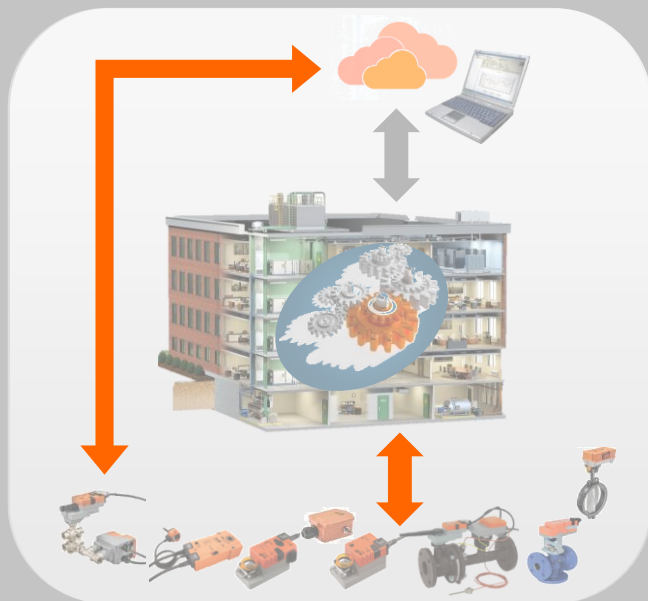
**Basics Networks and Field
Buses in Buildings**

Belimo Bus Portfolio

Lunch@Belgusto

Modbus RTU

Modbus@Belimo

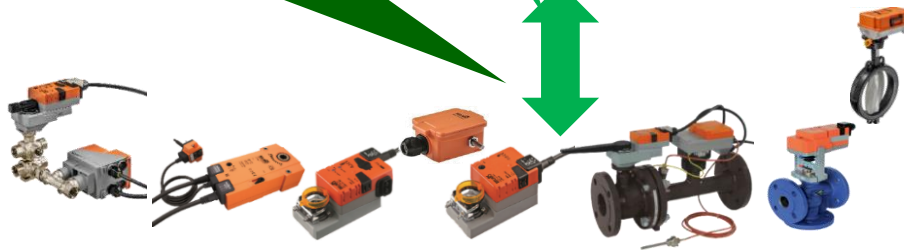


Modbus - History

**1979 Modbus:
Company Modicon
(Schneider Electric)**

**Takeover of many
manufacturers:
industry standard
(partially
standardized)**

DDC



**Industry,
Buildings, traffic**

**simple
protocol**

**efficiently
widely used**



Advantages of Joining | Join Form | Toolkit | Subscribe to Newsletter

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MODBUS NEWS

Modbus Security - A New Protocol to Improve Control System Security

The new protocol provides robust protection through the blending of Transport Layer Security (TLS) with the traditional Modbus protocol. Download a free copy of the specification at <http://modbus.org/tech.php>.
[Click here for more information...](#)

DISCUSSION FORUMS

[Create a Discussion](#)

Re: Need Help Interpreting MODBUS Documentation (reply posted by Carl_E)

Problem finding correct address in equipment documentation

[Need Help Interpreting](#)

MODBUS MEMBERS

a partial listing...

Danfoss



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PRODUCTS

CAS Gateway

TCP, Modbus

Weather Va

This new cus Gateway (CA in JSON paylo are extracted stored in an made availab IP, Modbus R 79 HTTP JSO injunction wi [Click here for](#)

Applied Mot

Motors with

Applications

Compact step are great for as medical de medical instr automation e camera contr

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Belimo develops, manufactures and distributes electric actuators for air dampers and valve technology for heating, ventilation and air conditioning systems since 1975. The technology company headquartered in Hinwil (Switzerland) employs more than 1100 people worldwide. Belimo products functionality and precision allow HVAC systems to achieve optimum performance and so reduce power consumption. Bus-capable actuators are equipped with intelligent actuator controls with open interfaces. This permits easy and inexpensive integration of air control and fire dampers, VAV units, control valves and intelligent HVAC performance devices.



What is Modbus?

De facto standard for industrial applications

- **open** (specifications on the Internet: www.modbus.org)
 - Modbus TCP part of IEC 61158 standard since 2007
- **License free**
- **Transmission bits or words (16 bits)** Manufacturer-independent

Modbus: Registers and Functions

Actuator

Modbus register:

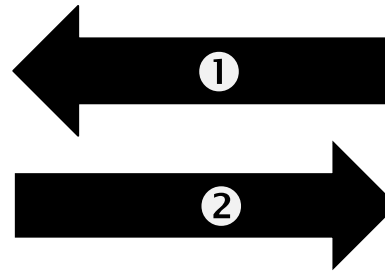
Rel. setpoint % (register no. 1)
Rel. position % (register no. 5)
Setpoint Source (Register No. 119)
...

**Modbus
SLAVE**

data
values

Modbus functions:

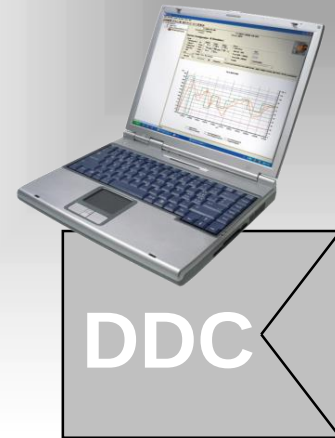
- Read Register value
- Write 1 Register value
- Write multiple registers



Send Register value

**Modbus
MASTER**

**Automation station
or
Management Station**



Modbus: Registers and Functions

Actuator

Modbus register:

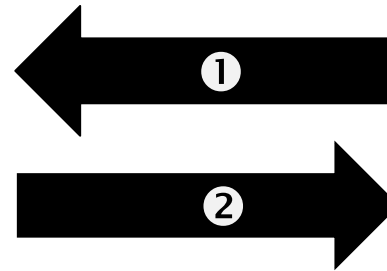
Rel. setpoint % (register no. 1)
Rel. position % (register no. 5)
Setpoint Source (Register No. 119)
...

Modbus
Slave

data
values

Modbus functions:

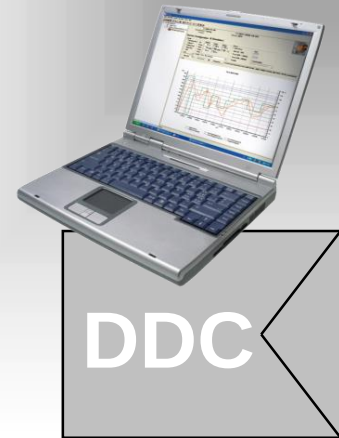
Example:
Write 1000 into register no. 1



Send Register value

Example: 1000
(001111101000 binary)

Automation station
or
Management Station



Modbus: Registers and Functions

Actuator

Modbus register:

Rel. setpoint % (register no. 1)

Rel. position % (register no. 5)

Setpoint Source (Register No. 119)

...

Modbus functions:

- Read Register value
- Write 1 Register value
- Write multiple registers

Register Description

Air Water Actuator		General actuator for air and water		BELIMO	
No.	Address	Description	Range	Unit	Scaling
1	0	Relative setpoint	0...100%	%	0.01
2	1	Relative position	0...100%	%	0.01
3	2	Setpoint source	0...100%	%	0.01
4	3	Actuator type	0...100%	%	0.01
5	4	Relative position	0...100%	%	0.01
6	5	Relative position	0...100%	%	0.01
7	6	Relative position	0...100%	%	0.01
8	7	Relative position	0...100%	%	0.01
9	8	Relative position	0...100%	%	0.01
10	9	Relative position	0...100%	%	0.01
11	10	Relative position	0...100%	%	0.01
12	11	Relative position	0...100%	%	0.01
13	12	Relative position	0...100%	%	0.01
14	13	Relative position	0...100%	%	0.01
15	14	Relative position	0...100%	%	0.01
16	15	Relative position	0...100%	%	0.01
17	16	Relative position	0...100%	%	0.01
18	17	Relative position	0...100%	%	0.01
19	18	Relative position	0...100%	%	0.01
20	19	Relative position	0...100%	%	0.01
21	20	Relative position	0...100%	%	0.01
22	21	Relative position	0...100%	%	0.01
23	22	Relative position	0...100%	%	0.01
24	23	Relative position	0...100%	%	0.01
25	24	Relative position	0...100%	%	0.01
26	25	Relative position	0...100%	%	0.01
27	26	Relative position	0...100%	%	0.01
28	27	Relative position	0...100%	%	0.01
29	28	Relative position	0...100%	%	0.01
30	29	Relative position	0...100%	%	0.01
31	30	Relative position	0...100%	%	0.01
32	31	Relative position	0...100%	%	0.01
33	32	Relative position	0...100%	%	0.01
34	33	Relative position	0...100%	%	0.01
35	34	Relative position	0...100%	%	0.01
36	35	Relative position	0...100%	%	0.01
37	36	Relative position	0...100%	%	0.01
38	37	Relative position	0...100%	%	0.01
39	38	Relative position	0...100%	%	0.01
40	39	Relative position	0...100%	%	0.01
41	40	Relative position	0...100%	%	0.01
42	41	Relative position	0...100%	%	0.01
43	42	Relative position	0...100%	%	0.01
44	43	Relative position	0...100%	%	0.01
45	44	Relative position	0...100%	%	0.01
46	45	Relative position	0...100%	%	0.01
47	46	Relative position	0...100%	%	0.01
48	47	Relative position	0...100%	%	0.01
49	48	Relative position	0...100%	%	0.01
50	49	Relative position	0...100%	%	0.01
51	50	Relative position	0...100%	%	0.01
52	51	Relative position	0...100%	%	0.01
53	52	Relative position	0...100%	%	0.01
54	53	Relative position	0...100%	%	0.01
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60	59	Relative position	0...100%	%	0.01
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65	64	Relative position	0...100%	%	0.01
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68	67	Relative position	0...100%	%	0.01
69	68	Relative position	0...100%	%	0.01
70	69	Relative position	0...100%	%	0.01
71	70	Relative position	0...100%	%	0.01
72	71	Relative position	0...100%	%	0.01
73	72	Relative position	0...100%	%	0.01
74	73	Relative position	0...100%	%	0.01
75	74	Relative position	0...100%	%	0.01
76	75	Relative position	0...100%	%	0.01
77	76	Relative position	0...100%	%	0.01
78	77	Relative position	0...100%	%	0.01
79	78	Relative position	0...100%	%	0.01
80	79	Relative position	0...100%	%	0.01
81	80	Relative position	0...100%	%	0.01
82	81	Relative position	0...100%	%	0.01
83	82	Relative position	0...100%	%	0.01
84	83	Relative position	0...100%	%	0.01
85	84	Relative position	0...100%	%	0.01
86	85	Relative position	0...100%	%	0.01
87	86	Relative position	0...100%	%	0.01
88	87	Relative position	0...100%	%	0.01
89	88	Relative position	0...100%	%	0.01
90	89	Relative position	0...100%	%	0.01
91	90	Relative position	0...100%	%	0.01
92	91	Relative position	0...100%	%	0.01
93	92	Relative position	0...100%	%	0.01
94	93	Relative position	0...100%	%	0.01
95	94	Relative position	0...100%	%	0.01
96	95	Relative position	0...100%	%	0.01
97	96	Relative position	0...100%	%	0.01
98	97	Relative position	0...100%	%	0.01
99	98	Relative position	0...100%	%	0.01
100	99	Relative position	0...100%	%	0.01
101	100	Relative position	0...100%	%	0.01
102	101	Relative position	0...100%	%	0.01
103	102	Relative position	0...100%	%	0.01
104	103	Relative position	0...100%	%	0.01
105	104	Relative position	0...100%	%	0.01
106	105	Relative position	0...100%	%	0.01
107	106	Relative position	0...100%	%	0.01
108	107	Relative position	0...100%	%	0.01
109	108	Relative position	0...100%	%	0.01
110	109	Relative position	0...100%	%	0.01
111	110	Relative position	0...100%	%	0.01
112	111	Relative position	0...100%	%	0.01
113	112	Relative position	0...100%	%	0.01
114	113	Relative position	0...100%	%	0.01
115	114	Relative position	0...100%	%	0.01
116	115	Relative position	0...100%	%	0.01
117	116	Relative position	0...100%	%	0.01
118	117	Relative position	0...100%	%	0.01
119	118	Relative position	0...100%	%	0.01
120	119	Relative position	0...100%	%	0.01
121	120	Relative position	0...100%	%	0.01
122	121	Relative position	0...100%	%	0.01
123	122	Relative position	0...100%	%	0.01
124	123	Relative position	0...100%	%	0.01
125	124	Relative position	0...100%	%	0.01
126	125	Relative position	0...100%	%	0.01
127	126	Relative position	0...100%	%	0.01
128	127	Relative position	0...100%	%	0.01
129	128	Relative position	0...100%	%	0.01
130	129	Relative position	0...100%	%	0.01
131	130	Relative position	0...100%	%	0.01
132	131	Relative position	0...100%	%	0.01
133	132	Relative position	0...100%	%	0.01
134	133	Relative position	0...100%	%	0.01
135	134	Relative position	0...100%	%	0.01
136	135	Relative position	0...100%	%	0.01
137	136	Relative position	0...100%	%	0.01
138	137	Relative position	0...100%	%	0.01
139	138	Relative position	0...100%	%	0.01
140	139	Relative position	0...100%	%	0.01
141	140	Relative position	0...100%	%	0.01
142	141	Relative position	0...100%	%	0.01
143	142	Relative position	0...100%	%	0.01
144	143	Relative position	0...100%	%	0.01
145	144	Relative position	0...100%	%	0.01
146	145	Relative position	0...100%	%	0.01
147	146	Relative position	0...100%	%	0.01
148	147	Relative position	0...100%	%	0.01
149	148	Relative position	0...100%	%	0.01
150	149	Relative position	0...100%	%	0.01

Modbus Slave

data values

Modbus: Register

Actuator

Modbus register:

Rel. setpoint % (register no. 1)

Rel. position % (register no. 5)

Setpoint Source (Register No. 119)

...

**Modbus
Slave**

**data
values**

Modbus functions:

- Read Register value
- Write 1 Register value
- Write multiple registers



Send Register value

**Modbus
MASTER**

**Automation station
or
Management Station**



DDC

Modbus Register: Operation

Example: NM24A MOD

Operation

No.	Address	Register	Access
1	0	Setpoint [%]	R / W
2	1	Override control	R / W
3	2	Command	R / W

Modbus Register Description

No.	Address	Description Comment	Range Enumeration	Unit	Scaling	Access
1	0	Setpoint Setpoint for actuator between Min (Register No. 106) and Max (No. 107)	0...10000 Default: 0	%	0.01	R / W
2	1	Override Control Override setpoint with defined values	0: None 1: Open 2: Close 3: Min 4: Mid 5: Max Default: None(0)	-	-	R / W
3	2	Command Initiation of actuator functions for service and test After command is sent, register returns to None(0)	0: None 1: Adaption 2: Test	-	-	R / W

Modbus Register Description

No.	Address	Description Comment	Range Enumeration	Unit	Scaling	Access
1	0	Setpoint Setpoint for actuator between Min (Register No. 106) and Max (No. 107)	0...10000 Default: 0	%	0.01	R / W

Write 1000 into Register 1 = Setpoint = 10%

Modbus Register: **Service** (Maintenance, Settings)

Example: NM24A MOD

Service

No.	Address	Register	Access
100	99	Bus termination	R
101	100	Series number 1 st part	R
102	101	Series number 2 nd part	
103	102	Series number 4 th part	
104	103	Firmware version	R
105	104	Malfunction and service information	R
106	105	Min [%]	R / W
107	106	Max [%]	R / W
108	107	Sensor type 1	R / W
109	108	Bus fail position	R / W
110	109	Communication Watchdog	R / W

for details see Firmware history						
105	104	Malfunction and Service Information Value is bit-coded. More than one bit can be set to 1 All bits not mentioned in the enumeration are not used for this actuator range Mechanical travel increased: The actuator has been moved outside the adapted working range Actuator cannot move: Mechanical overload e.g. blocked actuator, etc. Internal activity: Actuator performs a test run, adaption, etc. Gear disengaged: The gear disengaged button is pressed Bus Watchdog triggered: Timeout for the Bus Watchdog expired	Bit1: Mech travel increased Bit2: Actuator cannot move Bit8: Internal activity Bit9: Gear disengaged Bit10: Bus Watchdog triggered	–	–	R
106	105	Min	0...Max: 20%	%	0.01	R / W

Modbus Register: **Service** (Maintenance, Settings)

Example: NM24A MOD

Service

No.	Address	Register	Access
100	99	Bus termination	R
101	100	Series number 1 st part	R
102	101	Series number 2 nd part	
103	102	Series number 4 th part	
104	103	Firmware version	R
105	104	Malfunction and service information	R
106	105	Min [%]	R / W
107	106	Max [%]	R / W
108	107	Sensor type 1	R / W
109	108	Bus fail position	R / W
110	109	Communication Watchdog	R / W
111	110	–	–
112	111	–	–
113	112	–	–



108	107	Sensor 1 Type If Setpoint Source (Register 119) is analog (Hybrid mode) the sensor type 1 can be set to Active(1) in order to see the Setpoint Analog in mV	0: None 1: Active / Hybrid 2: Passive_1K 3: Passive_20K 4: Switch 5: PT1000_C 6: NI1000_C 7: NTC10K_C 8: PT1000_F 9: NI1000_F 10: NTC10K_F	–	–	R / W
-----	-----	---	---	---	---	-------

Modbus: Functions

Actuator

Modbus register:

Rel. setpoint % (register no. 1)

Rel. position % (register no. 5)

Setpoint Source (Register no. 9)

...

Modbus
Slave

data
values

Modbus functions:

- Read Register value
- Write 1 Register value
- Write multiple registers

1

Modbus

Automation station

Management Station



Commands

Standard commands:

Read Holding Registers [3]

Write Single Register [6]

Optional commands:

Read Discrete Inputs [2]

Read Input Registers [4]

Write Multiple Registers [16]

Modbus protocol

Language: Transport Options



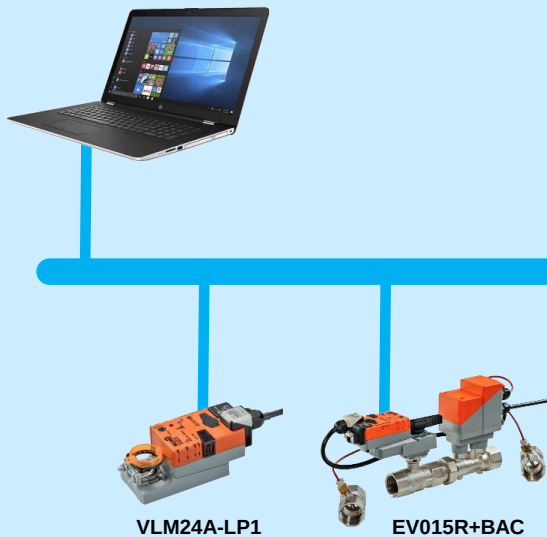
Transport options:

Modbus TCP
Modbus RTU
Modbus ASCII

....



Modbus TCP and Modbus RTU

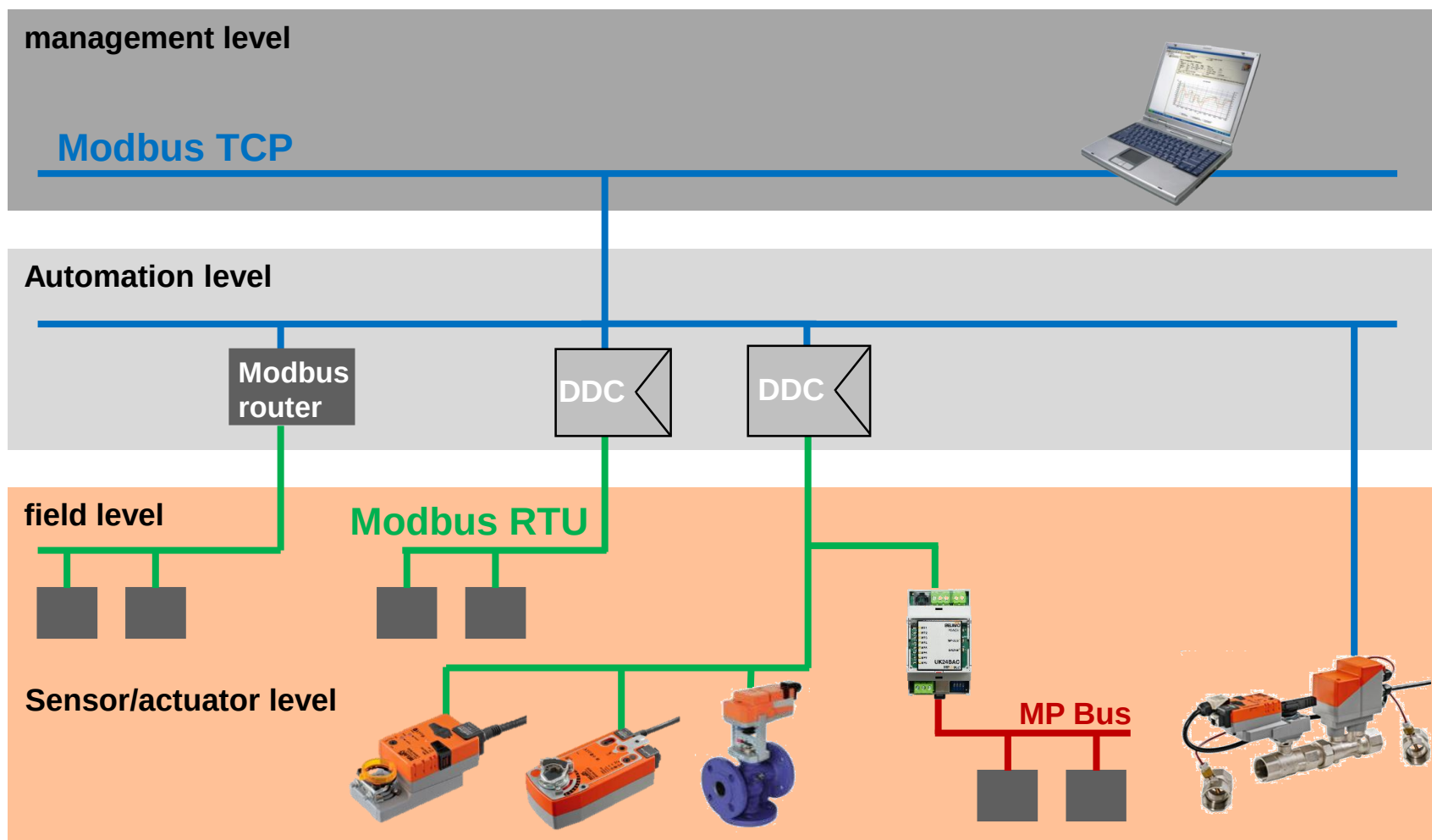


Modbus TCP:
IP Ethernet Communication



Modbus RTU:
RS-485 communication

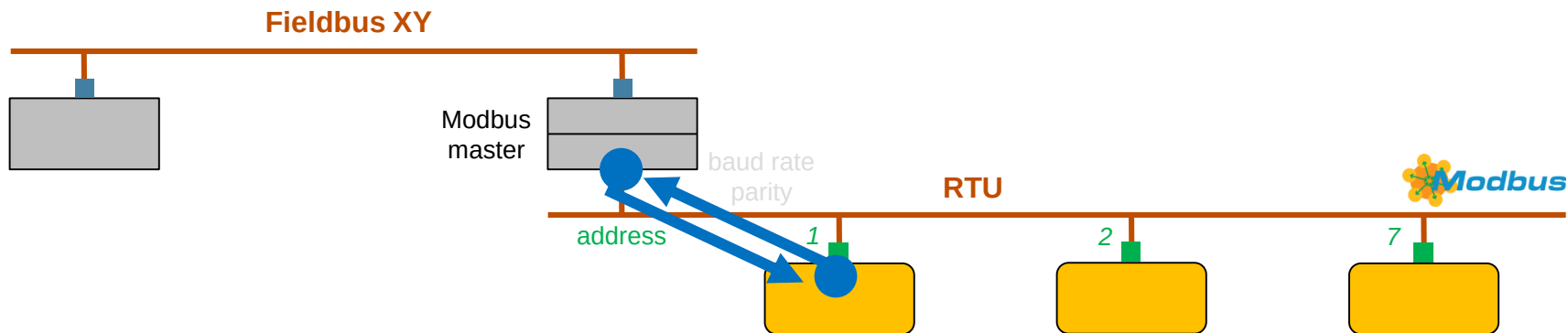
Modbus TCP and Modbus RTU



Modbus Address



- **1...247** Modbus address
- **0** is used for broadcasts

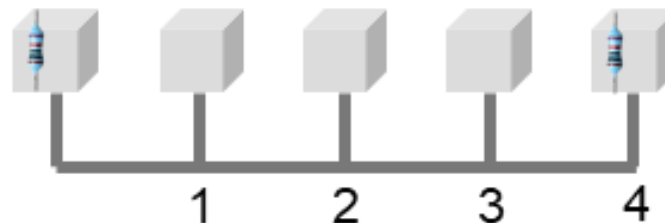


Address range: 1...247

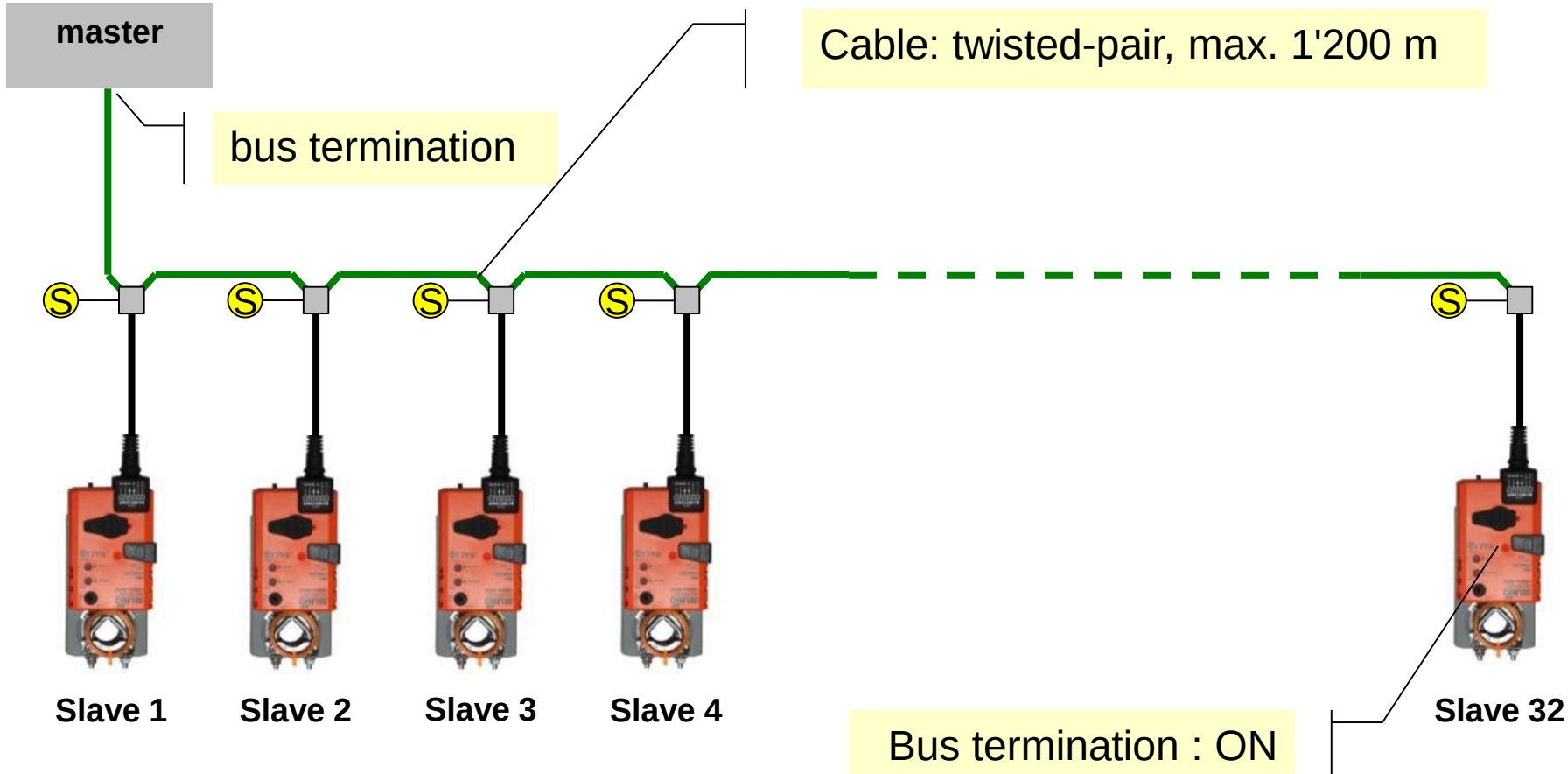
RS-485 Wiring

- Industry standard **EIA-485** for asynchronous data transmission
- Bus / line technology (Daisy Chain):
 - Stub lines generally not allowed
 - ⇒ **Therefore there are only short Actuator cable lengths for RS-485 communication such as Modbus RTU and BACnet MS/TP.**
- Longer cables must be terminated with a **terminating resistor** on both sides!
 - Install terminating resistor physically
 - if supported Switch on terminating resistor by software (e.g. with ZTH / App)

Bus line: Master Slave Slave Slave Slave



Bus line - classical topology



Power supply: AC/DC 24V, no galvanic isolation to Modbus

Translation: **MP** ⇔ **Modbus RTU Gateway**

Language translation / Protocol translation

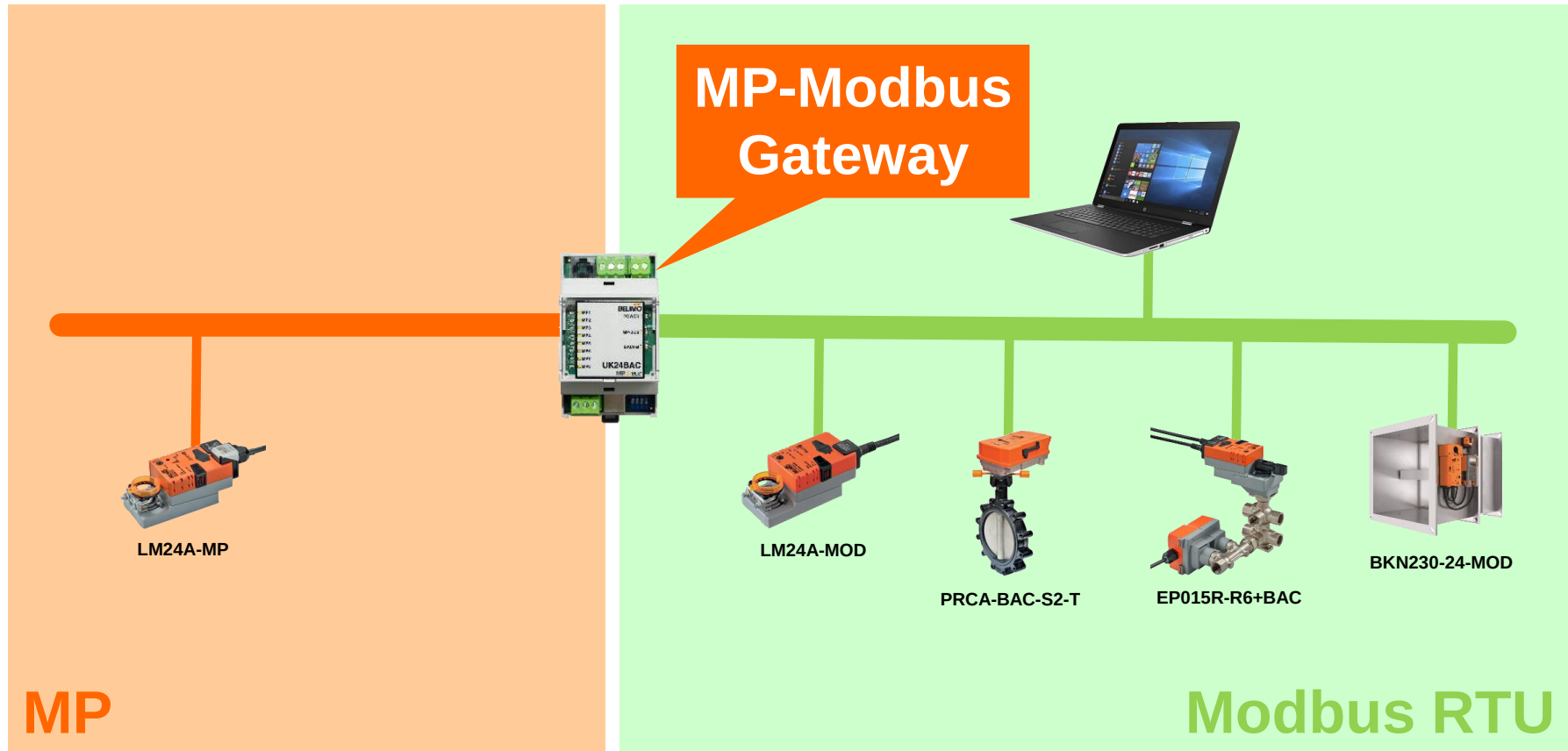


Gateway

UK24MOD:

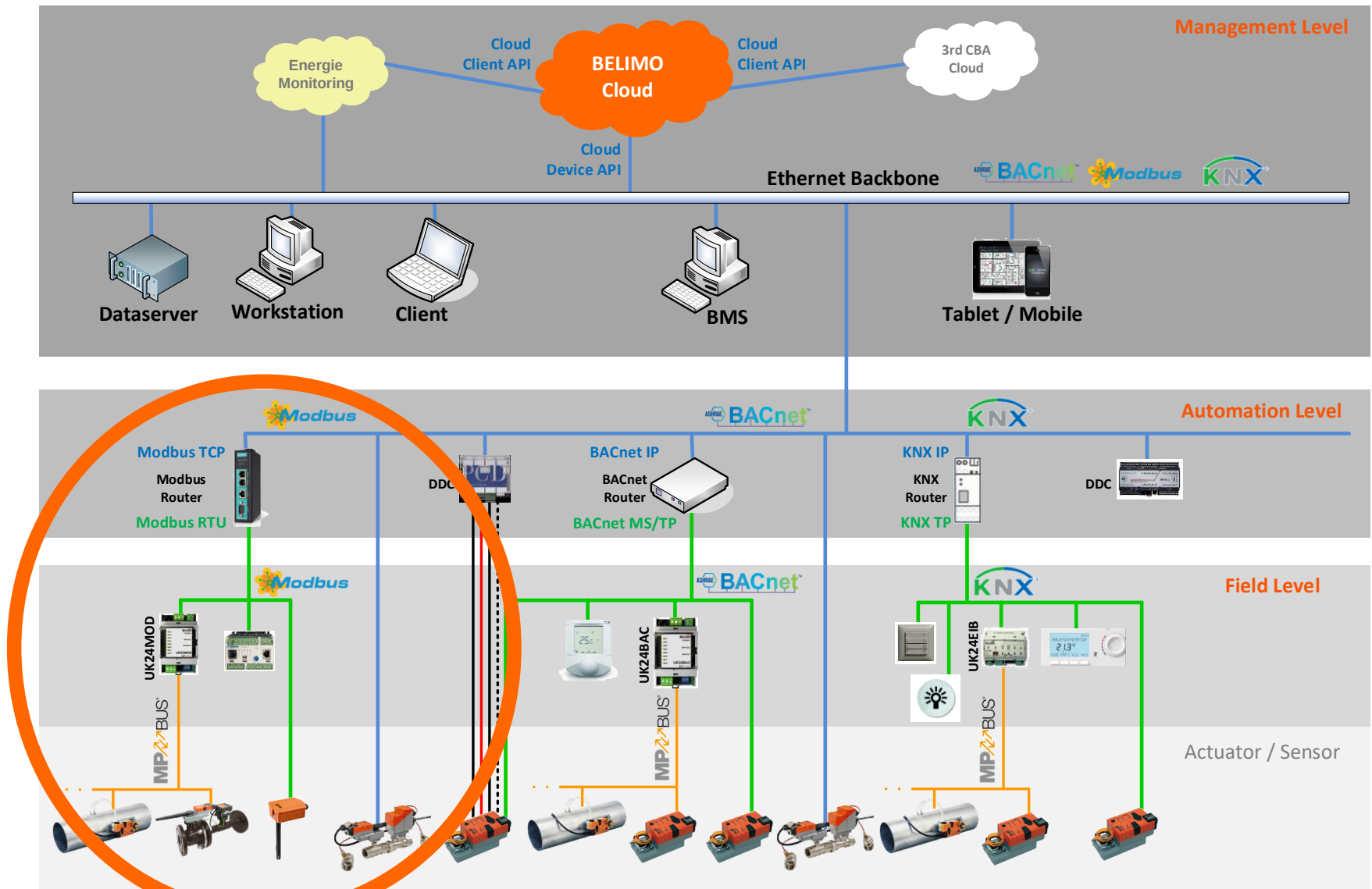
MP ⇔ Modbus RTU

Example Gateway



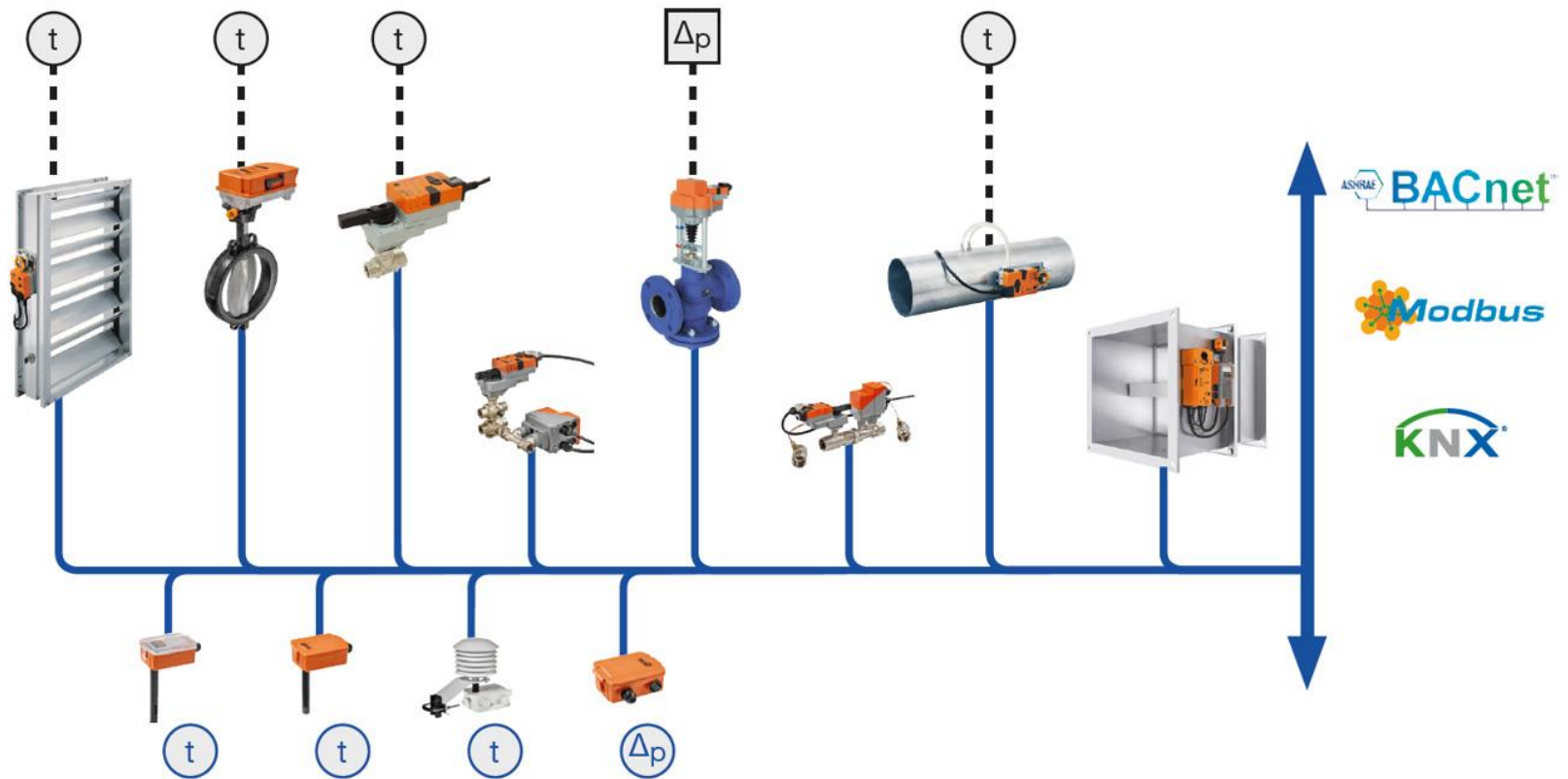
Attention: Not all actuators / sensors are supported by the gateways!

Modbus @Belimo



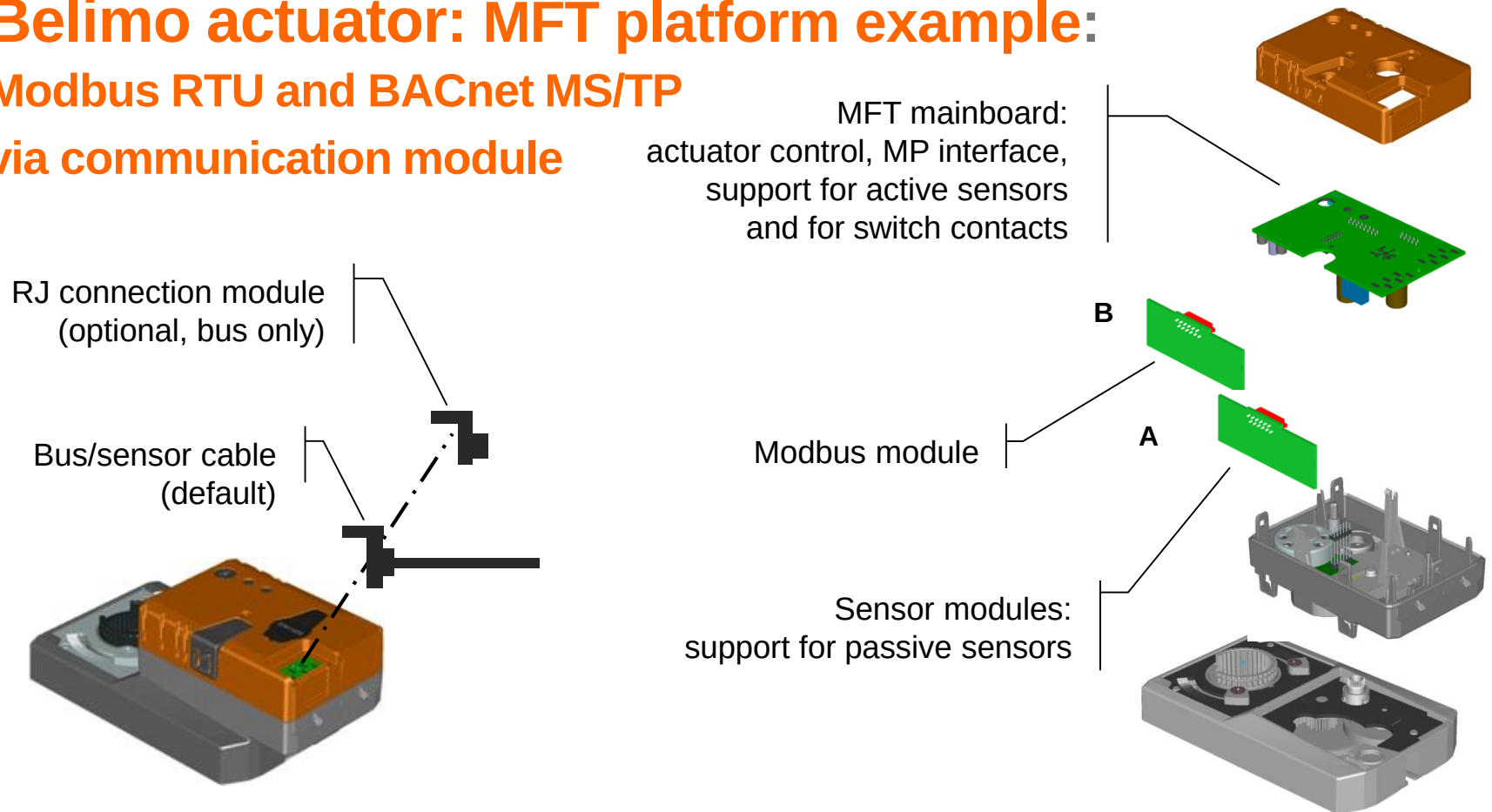
For direct system integrations

Modbus Portfolio



How does Modbus get on the Standard Actuators?

Belimo actuator: MFT platform example: Modbus RTU and BACnet MS/TP via communication module



Modbus@Belimo Product Portfolio



				
L...24A-MOD N...24A-MOD S...24A-MOD	SF24A-MOD	VAV-Compact	BKN230-24MOD	CQ24A-BAC
				
PRCA...-BAC	NV24A MOD	2-way EPIV	6-way EPIV	Energy Valve
				
Sensor Products	IoT Product Range		UK24MOD	

Modbus@Belimo Product Portfolio

Modbus TCP and Modbus RTU



L...24A-MOD N...24A-MOD S...24A-MOD	SF24A-MOD	VAV-Compact Modbus RTU	BKN230-24MOD	CQ24A-BAC
PRCA...-BAC	NV24A MOD Modbus RTU	2-way EPIV	6-way EPIV	Energy Valve Modbus TCP
Sensor Products RTU	IoT Product Range Modbus TCP		UK24MOD RTU	

Modbus@Belimo Product Portfolio

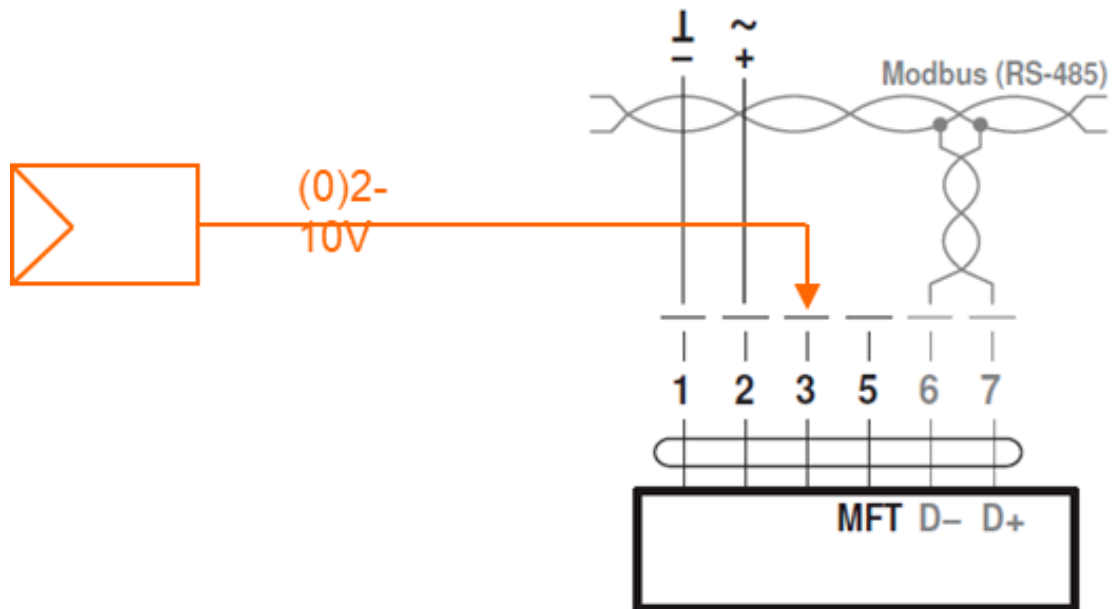


L...24A-MOD N...24A-MOD S...24A-MOD	SF24A-MOD	VAV-Compact	BKN230-24MOD	CQ24A-BAC
PRCA...-BAC	NV24A MOD	2-way EPIV	6-way EPIV	Energy Valve
Sensor Products	IoT Product Range	UK24MOD		

Portfolio Actuators Modbus / BACnet



- **Hybrid Mode** for Modbus and BACnet
 - Analog Control
 - Monitoring and override control via bus



Portfolio Actuators Modbus / BACnet

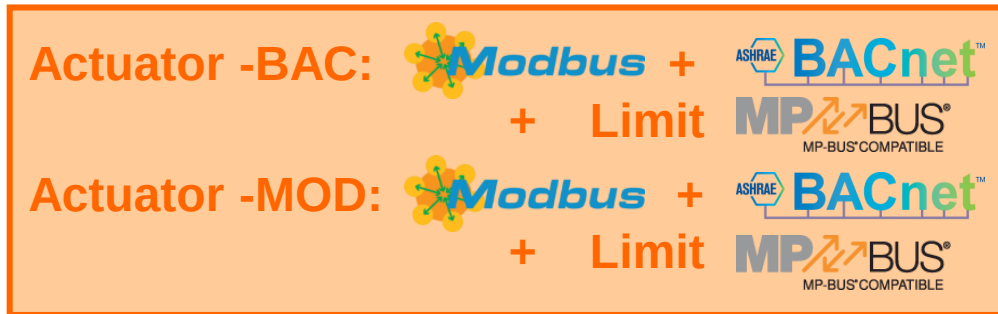
Bus Watchdog



Goal of the Bus Watchdog:

- **Check** whether Actuator is always **controlled (Setpoint periodically set)**
- If the actuator gets no setpoints-values anymore during **a specified amount of time**, then the Bus Watchdog is activated and the actuator goes to a **predefined position**.
- Available with Modbus and BACnet for many devices, Watchdog time configurable

Portfolio Actuators Modbus / BACnet



- Backwards compatible to older products with only 1 interface
- MP Bus supported with Limitation:
 - no COM LED for MP
 - no hybrid mode

Bus@Belimo Portfolio in Detail

Bus Brochure and Device Selection Tool



www.belimo.eu/busdevice

Device selection tool

Filter bus devices

Direct integration

- ☐ MP-Bus
- ☒ Modbus
 - ☒ RTU
 - ☐ TCP
- ☐ BACnet
- ☐ LON
- ☐ KNX
- ☐ Cloud

Sensor connection

- ☐ Active (0-10 V)
- ☐ Active (0-32 V)
- ☐ Passive (e.g. PT1000)
- ☐ Switching contact

Type

- ☒ Actuator
- ☐ Field device
- ☐ Sensor

Reset

MP-Bus with gateways

- ☐ via UK24MOD
- ☐ via UK24BAC
- ☐ via UK24LON
- ☐ via UK24EIB

Application

- ☐ Air
- ☐ Safety
- ☐ VAV
- ☐ Water

Fail-safe function

- ☐ with
- ☐ without

66 suitable devices

AVK24A-MOD

Globe v
AC/DC
Commu
MP-Bus
Running
Running
IP54, M

CQ24A-BAC

Rotary a
AC/DC
Commu
Running
IP40, M
Connec

EP015R+MOD

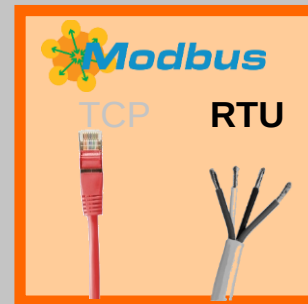
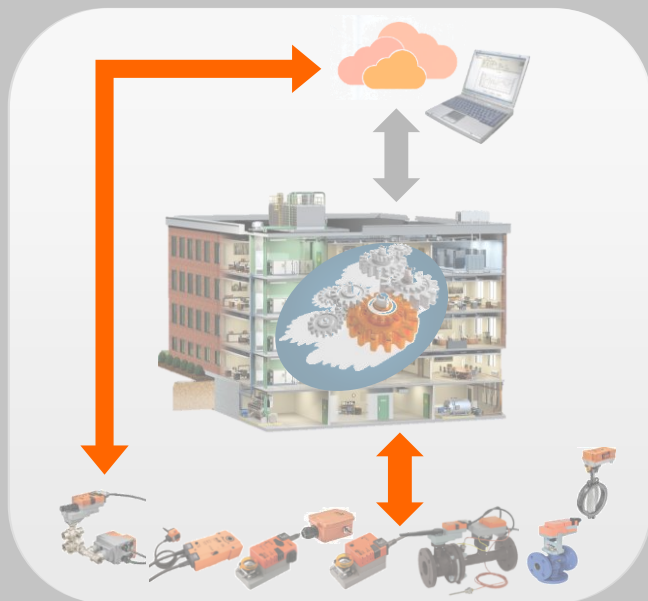
Electr. 2
Internal
AC/DC
Commu
MP-Bus
Adjusta
IP54, M
locked

EP015R-R6+BAC

Electr. 6
Internal
AC/DC
Commu

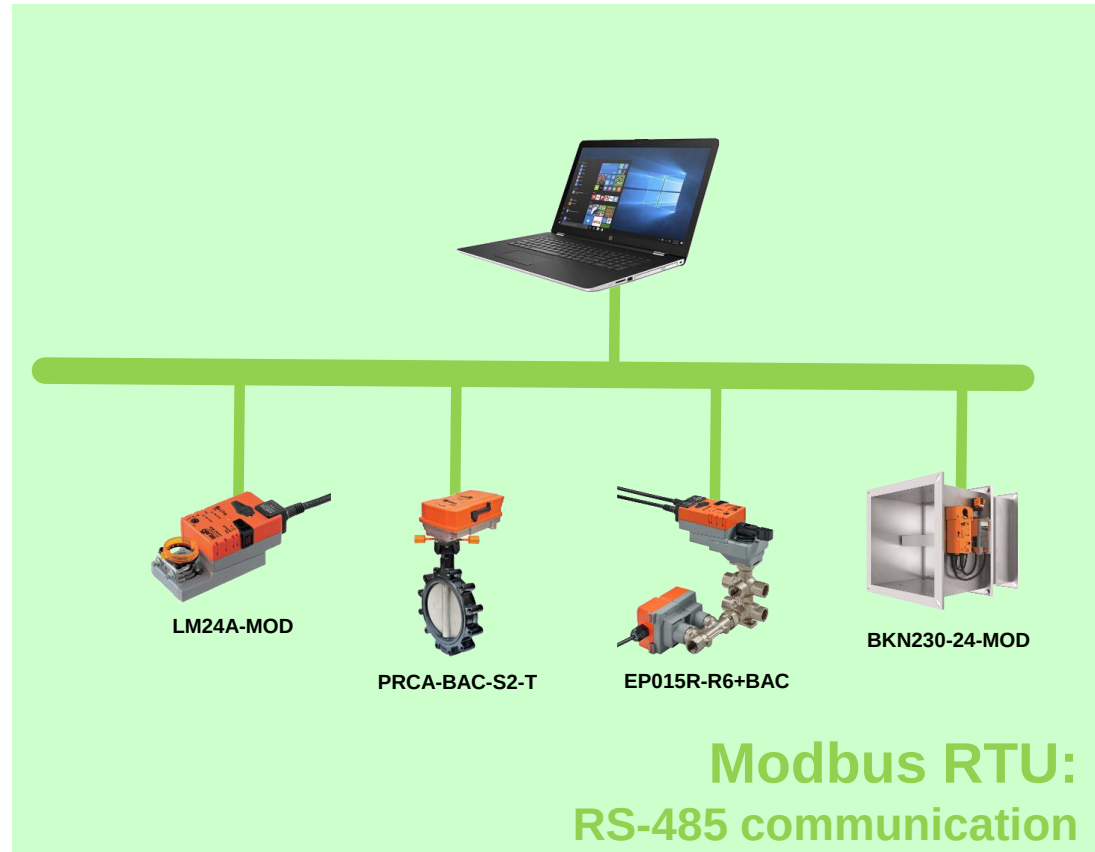


Modbus@Belimo

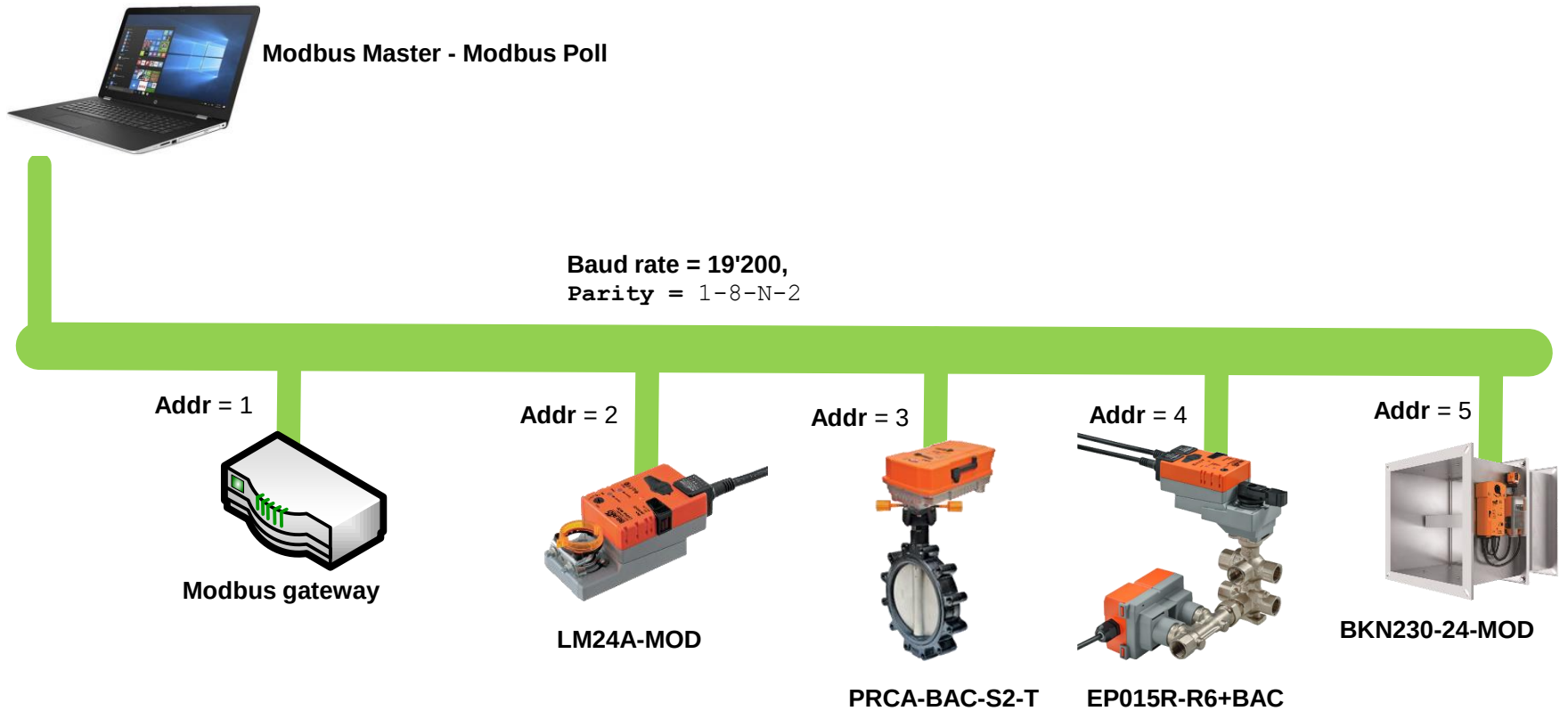


Modbus RTU

RS-485 Serial Communication



Modbus RTU Addressing



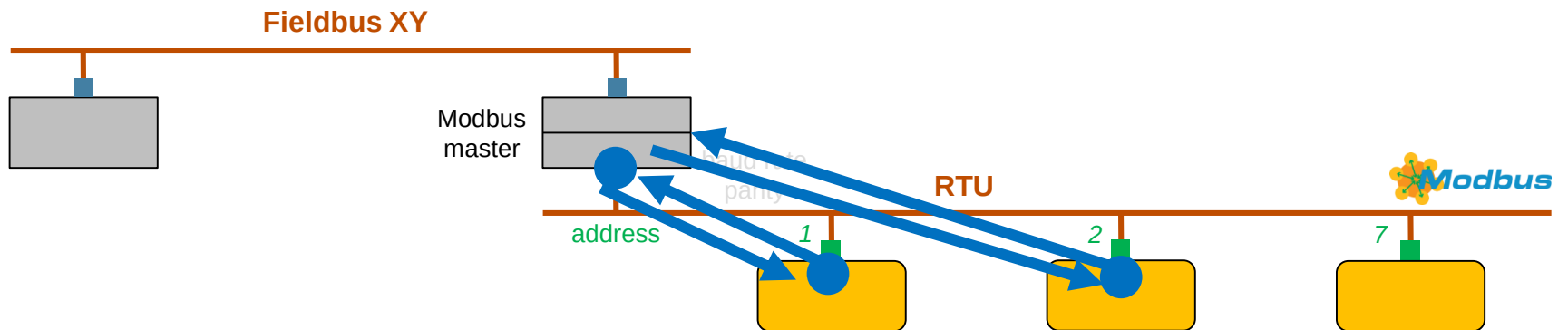
Parameterisation



Modbus Poll



Modbus RTU



Address range: 1...247

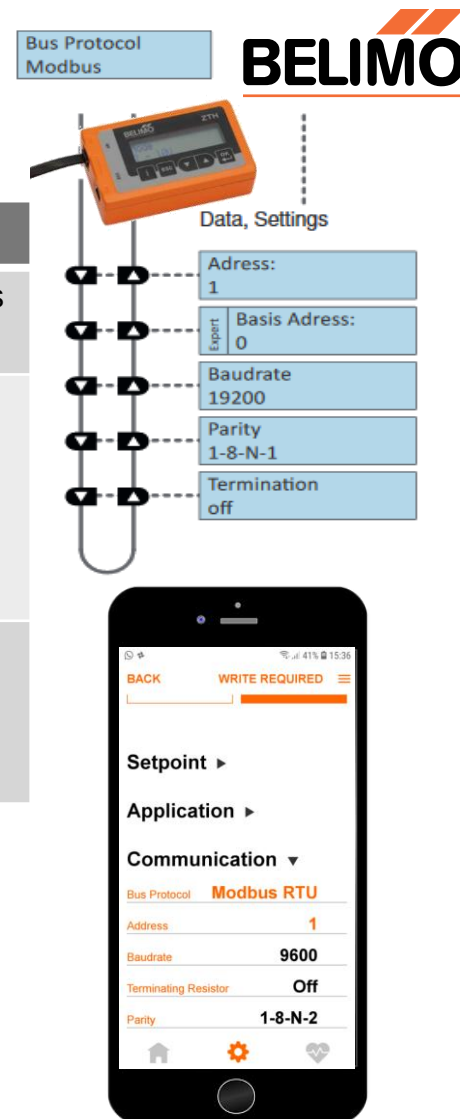
Modbus RTU - Configuration

Modbus RTU Configuration for Belimo Devices

Modbus Address	1-247	Modbus Address, unique in each Modbus Line
Baud rate	9 ' 600 19 ' 200 38 ' 400 76 ' 800 115 ' 200	Baud rate of the RS485 Network . Insert the Terminating Resistor at the first and last device of the line. Supported baud rates of the specific device can be found in the Register Description.
Parity	1-8-N-2 1-8-N-1 1-8-E-1 1-8-O-1	Transmission format must be set the same on the whole bus line, including the master and all slaves.



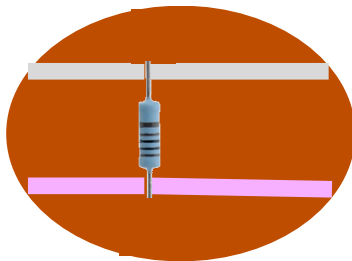
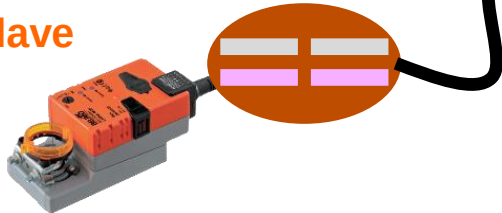
Check max. number of Devices per Modbus Line (typically 32 Devices)



Workshop Modbus Poll and Wiring

Modbus Master - Modbus Poll

Modbus
slave



USB

The image shows two screenshots of the Modbus Poll software interface. The left screenshot shows the 'Register 1_10.mbp' window, and the right screenshot shows the 'Register 100_109.mbp' window. Both windows display a table of registers with their addresses, aliases, and values.

Address	Alias	Value
0	-	65535
1	Zwangssteuerung	1
2	Kommando	0
3	Antriebstyp	3
4	Relative Position [%]	10000
5	Aktuelle Leistung [mW]	900
6	-	65535
7	-	65535
8	Sammelstörung	0
9	-	-

Address	Alias	Value
0	Series Number 1st	0
1	Series Number 2nd	0
2	Series Number 4th	0
3	Firmware	207
4	Maintenance	0
5	-	65535
6	-	65535
7	Busausfallzeit	0
8	Bus Fail Position	0
9	Communication Watchdog	300

For communication problems:
120 ohm resistance between pink&grey

Modbus RTU

Raw data in Modbus Poll

The image shows two side-by-side windows from the Modbus Poll software. The left window is titled 'Register 1_10.mbp' and the right window is titled 'Register 100_109.mbp'. Both windows display a table of data with columns for 'Alias' and a numerical value. The status bar at the top of each window shows 'Tx = 57: Err = 0: ID = 1: F = 03: SR = 1000ms' for the left and 'Tx = 53: Err = 0: ID = 1: F = 03: SR = 1000ms' for the right.

	Alias	Value
0		65535
1		1
2		0
3		3
4		10000
5		900
6		65535
7		65535
8		0
9		

	Alias	Value
0		0
1		0
2		0
3		207
4		0
5		65535
6		65535
7		0
8		0
9		300

Binary system:

<https://www.youtube.com/watch?v=LpuPe81bc2w&t=252s> (6'00")

Any questions?





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